

Final model recipe

Parameter	Selected recipe
Backbone	Midnight-12k
Target crop	12 μm / 57 native px at 0.2125 μm/px
Input and feature	224 × 224; CLS + mean patch tokens; 3,072 dim
Trainable weights	Final transformer block + 3,072 → 7 head
Training	3 epochs; AdamW; weighted cross entropy
Learning rates	Backbone 1e-5; head 1e-3; weight decay 1e-4
Augmentation	90° rotations, horizontal flip, mild brightness/contrast
Seed	20260923
Performance source	Corrected five-fold grouped cross-validation
Deployment	Independent slide/patient validation still required